

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks: 30

Annexure A

Resolution Algorithm Implementation

Code of Conduct: (19241091)

I POOJASREE B (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Poojasree B

$20 + 20 + 20 + 18 + 13$
 $= 91$

RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	20
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	20
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	23
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	18
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	13

= 94

10

ARTIFICIAL INTELLIGENCE

CD3_AT3

POOJASREE B

192411091

CSA716

1. Car Rental Eligibility

1. Propositional Logic Representation

Let:

V = Suresh has a valid license

R = Suresh can rent a car

Rule: $V \rightarrow R$

Fact: V

Goal: R

2. Convert to CNF and Negate Goal

Rule: $V \rightarrow R$

Eliminate implication: $\neg V \vee R$

Fact: V

Negated goal: $\neg R$

So the clauses are:

1. $\neg V \vee R$

2. V

3. $\neg R$

3. Resolution Steps

Step 1

Resolve: $(\neg V \vee R)$

with: V

The complementary literals V and $\neg V$ cancel.

Therefore: R

Step 2

Resolve: R

with the negated goal: $\neg R$

We get: ϕ

4. Empty clause

$R, \neg R$

$\therefore \phi$

The empty clause represents a contradiction.

Final conclusion

Since the negation of the goal leads to a contradiction:

Suresh can rent a car.

Thus, the goal is proved using the Resolution Algorithm.

2. Sports Achievement Award

1. Propositional Logic

Let:

P = Divya scored above 90%.

G = Divya wins a gold medal

Rule: $P \rightarrow G$

Fact: P

Goal: G

2. CNF and Negated Goal

Rule: $P \rightarrow G$

CNF: $\neg P \vee G$

Fact: P

Negated goal: $\neg G$

3. Resolution Steps

Step 1

Resolve: $\neg P \vee G$

with: P

Therefore: G

Step 2

Resolve: G

with the negated goal: $\neg G$

Therefore: ϕ

4. Empty clause

$G, \neg G$

$\therefore \phi$

Since the negation of the goal produces a contradiction:

Divya wins a gold medal.

Thus, the goal is proved using the Resolution Algorithm.

3. Job Interview Outcome

1. Represent in propositional logic

Let:

P = Farhan passed the interview

H = Farhan is hired

Knowledge Base:

1. If an applicant passes the interview, then

the applicant is hired.

$$P \rightarrow H$$

2. Farhan passed the interview.

P

Goal: H

2. Convert to CNF and Negate the Goal

Statement 1

$$P \rightarrow H$$

Remove implication:

$$\neg P \vee H$$

So the first clause is:

$$C_1 = \{\neg P, H\}$$

Statement 2

P

So:

$$C_2 = \{P\}$$

Negate the goal

Goal: H

Negated goal: $\neg H$

$$\text{So: } C_3 = \{\neg H\}$$

Therefore, the clauses are:

Clause	Expression
--------	------------

C_1	$\neg P \vee H$
-------	-----------------

C_2	P
-------	-----

C_3	$\neg H$
-------	----------

3. Apply Resolution Algorithm

Step 1

Resolve C_1 and C_2 :

$(\neg P \vee H), P$

The complementary literals are:

$\neg P$ and P

Cancel them:

H

Therefore:

$C_4 = \{H\}$

Step 2

Resolve C_4 with C_3 :

$H, \neg H$

The complementary literals are:

H and $\neg H$

Cancel them: ϕ

The empty clause has been obtained.

4. Resolution Proof

You can write the final answer like this in your assignment:

$$C_1: \neg P \vee H$$

$$C_2: P$$

$$C_3: \neg H \text{ (Negated Goal)}$$

Resolution:

$$C_1 (\neg P \vee H) + C_2 (P) \\ \Rightarrow H$$

Then:

$$H + \neg H \Rightarrow \phi$$

Conclusion

Since resolution produces the empty clause ϕ , the negation of the goal leads to a contradiction.

Therefore,

Farhan is hired.

Hence, the goal is proved using the Resolution Algorithm.

4. Attendance Credit

1. Proposition Logic

Let:

A = Meena attended all classes

C = Meena gets attendance credit

Knowledge Base:

1. If a student attends all classes, then the student gets attendance credit.

$$A \rightarrow C$$

2. Meena attended all classes.

A

Goal:

C

2. Convert to CNF and Negate the Goal

Statement 1

$$A \rightarrow C$$

Remove the implication:

$$\neg A \vee C$$

Therefore:

$$C_1 = \{\neg A, C\}$$

Statement 2

A

Therefore:

$$C_2 = \{A\}$$

Negated Goal

Goal: C

Negate it: $\neg C$

Therefore:

$$C_3 = \{\neg C\}$$

So the clause set is:

Clause	Expression
--------	------------

C_1	$\neg A \vee C$
-------	-----------------

C_2	A
-------	---

C_3	$\neg C$
-------	----------

3. Apply Resolution Algorithm

Step 1: Resolve C_1 and C_2

$$(\neg A \vee C), A$$

The complementary literals are $\neg A$ and A

Therefore: $C_4 = \{C\}$

So we get: C

Step 2: Resolve C_4 and C_3

$$C, \neg C$$

The complementary literals are C and $\neg C$

Therefore: ϕ

The empty clause is obtained.

4. Final Resolution Proof

$$C_1: \neg A \vee C$$

$$C_2: A$$

$$C_3: \neg C$$

$$\text{Resolution: } C_1 + C_2 \Rightarrow C$$

$$\text{Then: } C + C_3 \Rightarrow \phi$$

Conclusion

Since the empty clause ϕ is derived, the negation of the goal leads to a contradiction.

$\therefore$ Meena gets attendance credit.

5. Order Shipping Confirmation

1. Represent the knowledge Base in Propositional Logic

Let:

P = Payment is Verified

C = Order is confirmed

S = Item is shipped

Knowledge Base:

1. If payment is verified, then the order is confirmed.

$$P \rightarrow C$$

2. If the order is confirmed, then the item is shipped.

$$C \rightarrow S$$

3. The payment was verified.

$$P$$

Goal: S

2. Convert to CNF

Statement 1

$$P \rightarrow C$$

$$\neg P \vee C$$

$$\text{Therefore: } C_1 = \{\neg P, C\}$$

Statement 2

$$C \rightarrow S$$

$$\neg C \vee S$$

$$\text{Therefore: } C_2 = \{\neg C, S\}$$

Statement 3

$$P$$

$$\text{Therefore: } C_3 = \{P\}$$

3. Negate the Goal

Goal: S

Negated goal: $\neg S$

$$\text{Therefore: } C_4 = \{\neg S\}$$

4. Apply Resolution Algorithm

Step 1: Resolve C_1 and C_3

$(\neg P \vee C), P$

The complementary literals are $\neg P$ and P .

Therefore: C

So, $C_5 = \{C\}$

This means the order is confirmed.

Step 2: Resolve C_2 and C_5

$(\neg C \vee S), C$

The complementary literals are $\neg C$ and C

Therefore: S

So, $C_6 = \{S\}$

This means the item is shipped.

Step 3: Resolve C_6 and C_4

$S, \neg S$

The complementary literals are S and $\neg S$

Therefore: $\emptyset$

The empty clause is obtained.

5. Final Resolution Proof

$C_1 : \neg P \vee C$

$C_2 : \neg C \vee S$

$C_3 : P$

$$C_4: \neg S$$

Resolution steps:

$$C_1 + C_3 \Rightarrow C$$

$$C_2 + C \Rightarrow S$$

$$S + C_4 \Rightarrow \phi$$

Conclusion

Since the empty clause ϕ is derived,
the negation of the goal is contradictory.

$\therefore$ The item is shipped.